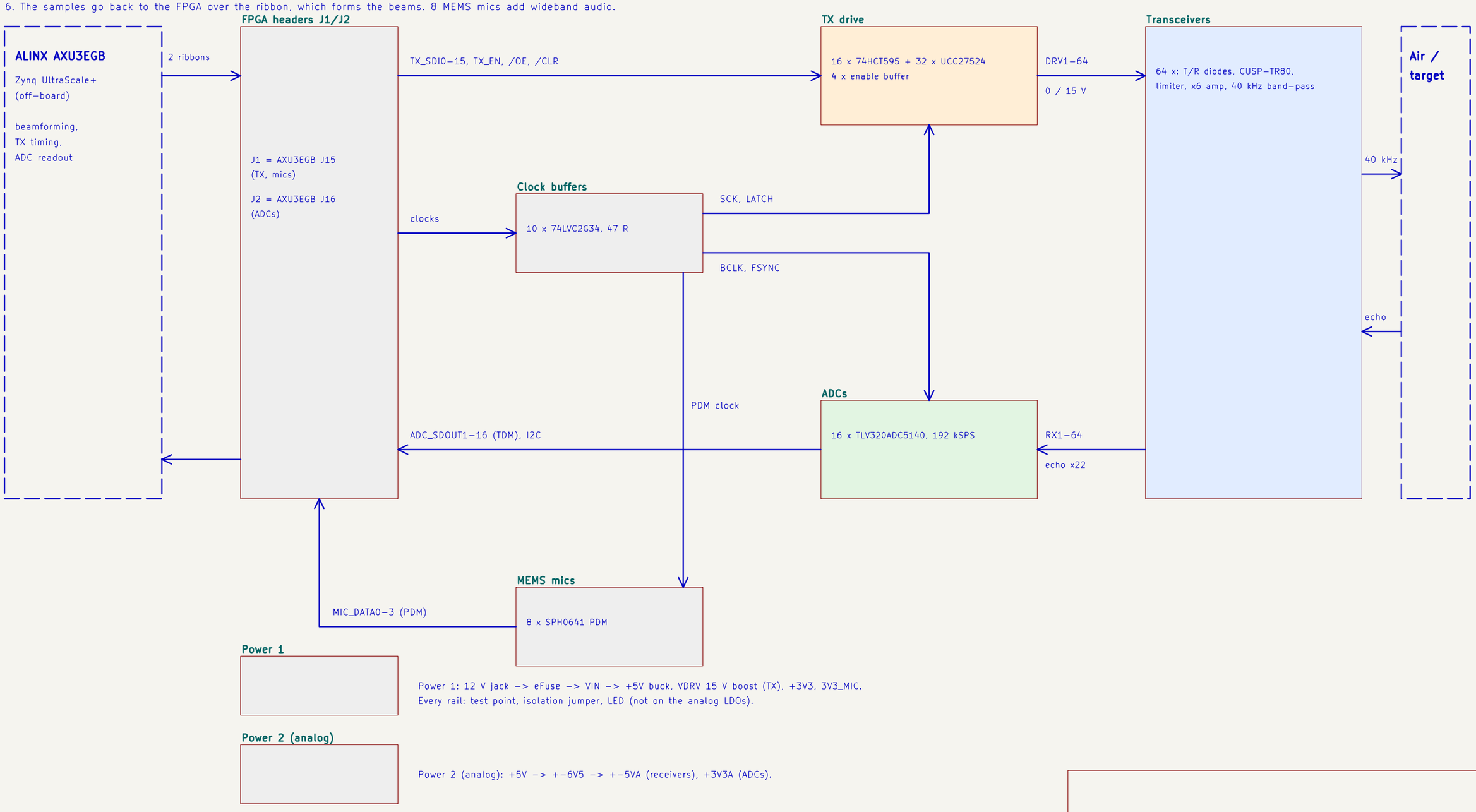
Hybrid 8x8 ultrasonic AESA: how the signal flows

1. The FPGA streams one TX bit per element (16 serial lines, 10 MHz) and the clocks.
2. TX drive: shift registers latch the bits; gate drivers switch each element between 0 and 15 V.
3. Transceivers: the 64 CUSP-TR80 elements emit 40 kHz; echoes return to the same elements.
4. The diode T/R switch and limiter protect the receiver; amplifier + 40 kHz band-pass give x22.
5. ADCs: 16 x TLV320ADC5140 sample all 64 channels at once, 192 kSPS (raw waveform, amplitude and phase).
6. The samples go back to the FPGA over the ribbon, which forms the beams. 8 MEMS mics add wideband audio.



Power 1: 12 V jack -> eFuse -> VIN -> +5V buck, VDRV 15 V boost (TX), +3V3, 3V3_MIC.
Every rail: test point, isolation jumper, LED (not on the analog LDOs).

Power 2 (analog): +5V -> +-6V5 -> +-5VA (receivers), +3V3A (ADCs).

Click a block to open it. Repeated parts are one sheet each: channel x64, adc x16, tx_group x16, en_buf x4, clk_buf x10, mic x8.

Hybrid 8x8 ultrasonic AESA	
Sheet: /	
File: aesa.kicad_sch	
Title: Hybrid 8x8 ultrasonic AESA: block diagram	
Size: A3	Date: 2026-10-24
KiCad E.D.A. 10.0.6	Id: 1/15

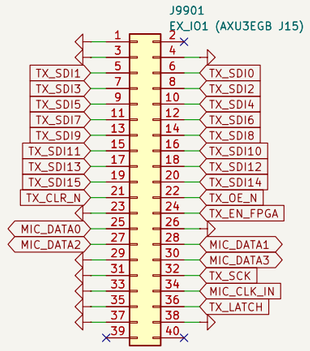
Rev: 1

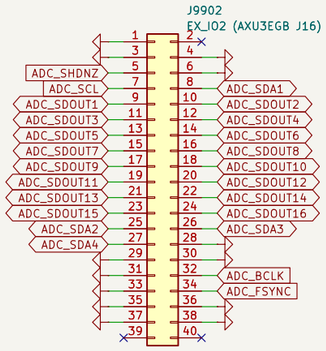
<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

AXU3EGB expansion headers (40-pin IDC, 2.54 mm). J1 = board J15 (TX, mics), J2 = board J16 (ADCs: 16 TDM lines, BCLK, FSYNC, I2C).

Pin 2 (+5V) and 39/40 (+3.3V) of the FPGA board are left open. Clocks sit between GND conductors (pins 31–37).

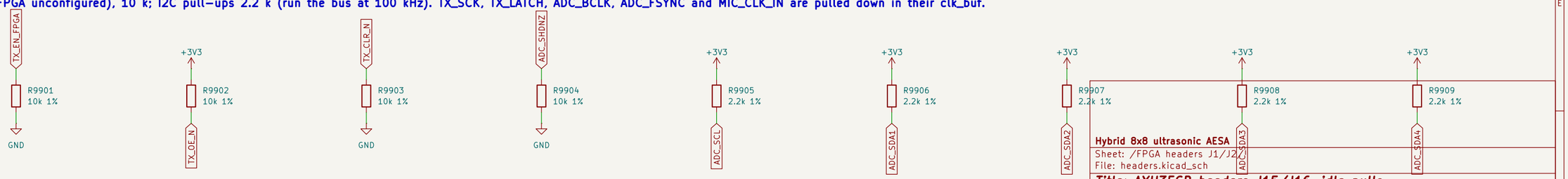
Spare FPGA pins tied to GND must stay unused (inputs) in the XDC.





FPGA ribbon outputs: DRIVE 4, SLEW SLOW (near source-matched to the 100-ohm ribbon, sim/reports/clock_si_drive.md): no termination parts needed.

Idle pulls (FPGA unconfigured), 10 k; I2C pull-ups 2.2 k (run the bus at 100 kHz). TX_SCK, TX_LATCH, ADC_BCLK, ADC_FSYNC and MIC_CLK_IN are pulled down in their clk_buf.



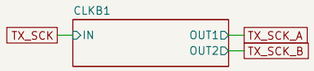
Hybrid 8x8 ultrasonic AESA
Sheet: /FPGA headers J1/J2/
File: headers.kicad_sch

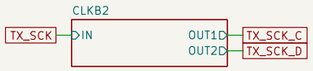
Title: AXU3EGB headers J15/J16, idle pulls

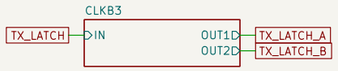
Size: A3	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 2/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

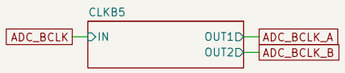
Clock buffers. Each FPGA clock enters one or two clk_buf cells; every output drives at most 4 loads through 47 R.

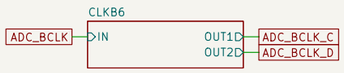


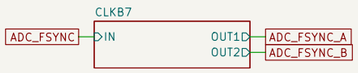


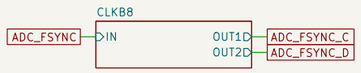
















Hybrid 8x8 ultrasonic AESA

Sheet: /Clock buffers/
File: clocks.kicad_sch

Title: Clock buffers: 10 x 74LVC2G34

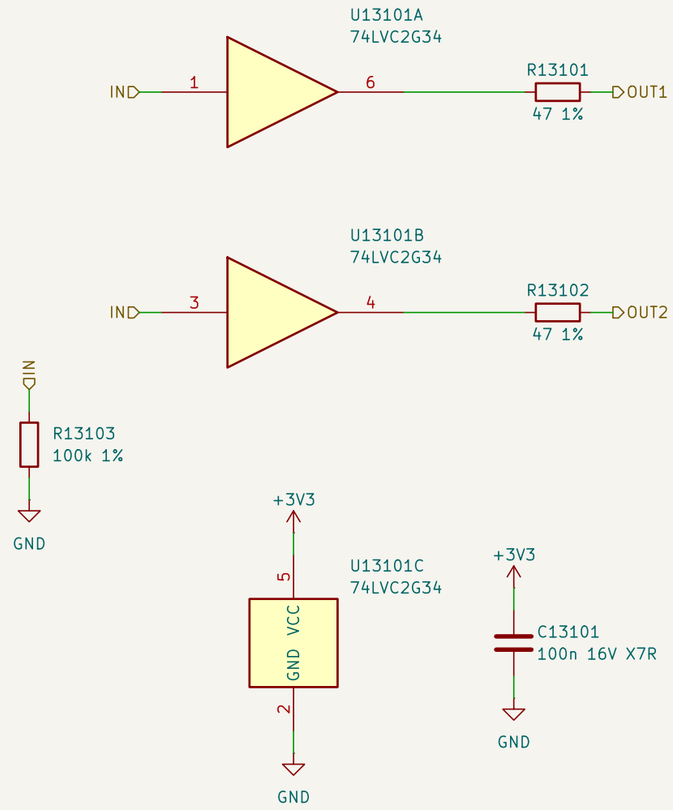
Size: A3	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 3/15

<< up to: Clock buffers: 10 x 74LVC2G34 This sheet is used 10 times; the page shows the first instance.

Clock fan-out: 74LVC2G34, both gates on IN, 47 R series termination per output (with the ~20 R driver: matches the 61 ohm 0.25 mm clock traces)

Each output drives at most 4 loads, clustered at the far end of its trace (series termination needs that: sim/reports/mic_clock_si.md).

100 k pull-down keeps the clock low while the FPGA is unconfigured.



Hybrid 8x8 ultrasonic AESA

Sheet: /Clock buffers/CLKB1/

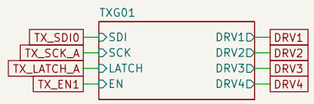
File: clk_buf.kicad_sch

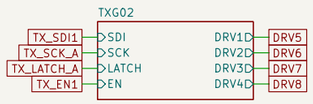
Title: Clock buffer (x10): 74LVC2G34, 47 R outputs

Size: A4	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 4/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

TX drive. The FPGA streams 16 serial lines (TX_SDI0–15) into 16 tx_group cells; each drives 4 elements (DRV). TX_EN_FPGA reaches the drivers through 4 en_buf cells.





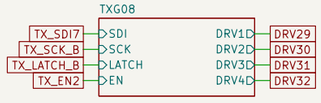


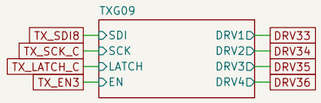




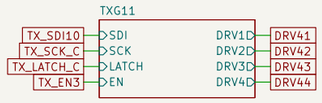


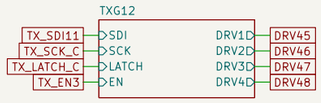














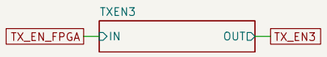














Hybrid 8x8 ultrasonic AESA

Sheet: /TX drive/
File: tx.kicad_sch

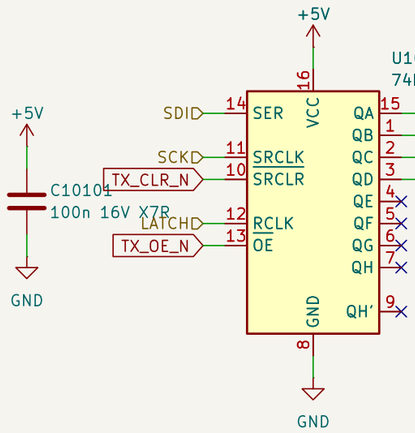
Title: TX drive: 16 shift–register/driver groups, 4 enable buffers

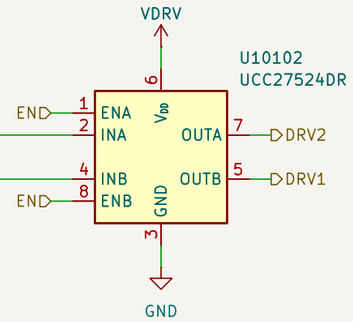
Size: A3	Date: 2026–10–24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 5/15

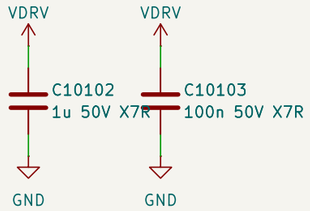
<< up to: TX drive: 16 shift-register/driver groups, 4 enable buffers

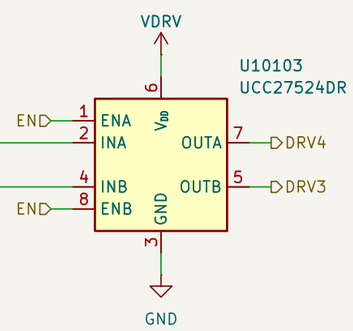
TX group: 4 elements. 74HCT595 QA–QD -> UCC27524 inputs; QE–QH unused (16 data lines, 10 MHz SRCLK)

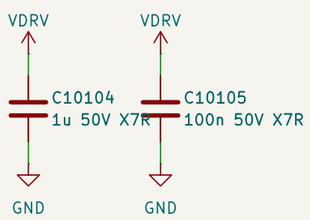
EN from one 74AHCT1G125 per 8 drivers (en_buf). SCK/LATCH from a clk_buf output (<= 4 loads).











Hybrid 8x8 ultrasonic AESA

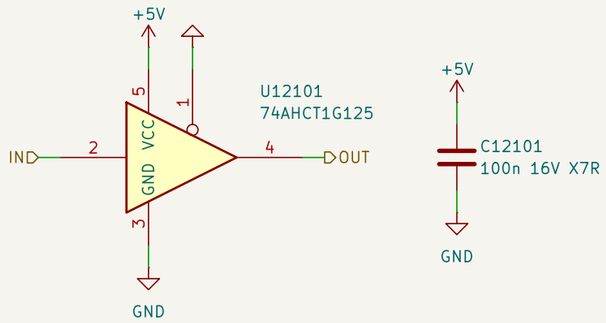
Sheet: /TX drive/TXG01/

File: tx_group.kicad_sch

Title: TX group (x16): 74HCT595 + 2 x UCC27524, 4 elements

Size: A4	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 6/15

TX enable buffer: 74AHCT1G125, 3.3 V in (TTL threshold), 5 V out to 8 UCC27524 (16 EN pins)



Hybrid 8x8 ultrasonic AESA

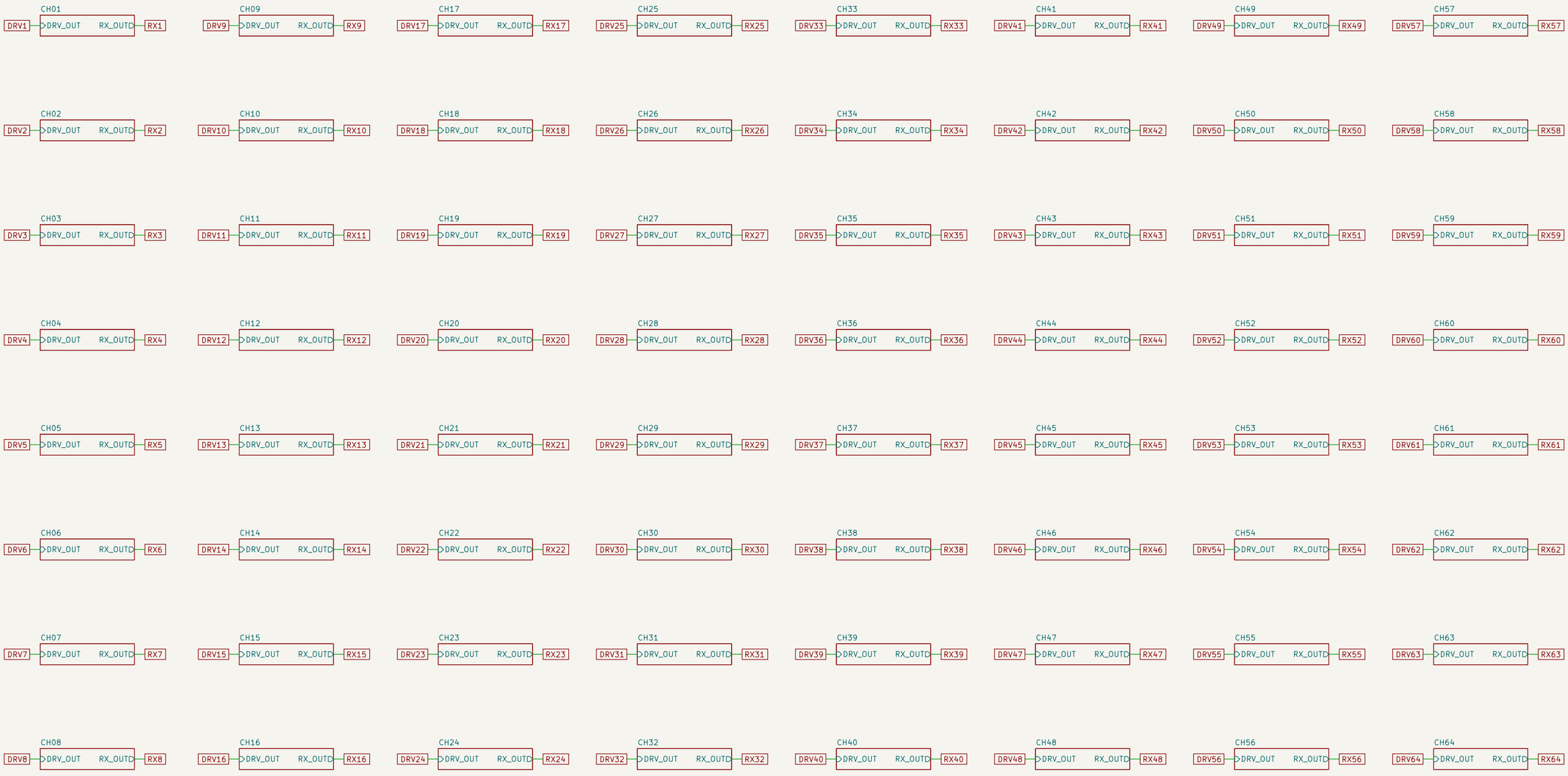
Sheet: /TX drive/TXEN1/
File: en_buf.kicad_sch

Title: TX enable buffer (x4): 74AHCT1G125

Size: A4	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 7/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

Transceiver array. Channel n = element n of the 8 x 8 grid, numbered down the columns (as on the board). Each channel takes DRVn from the TX drive and returns its amplified echo RXn to the ADCs.



Hybrid 8x8 ultrasonic AESA

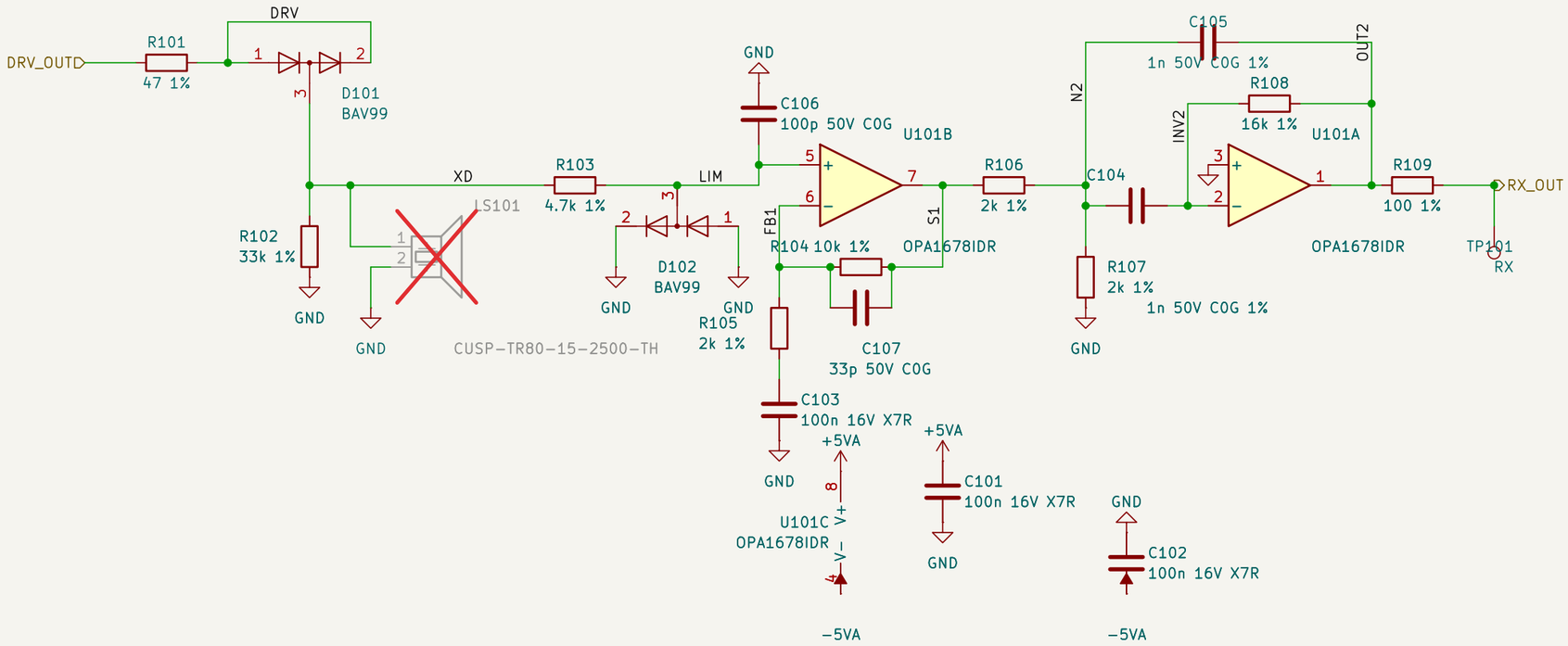
Sheet: /Transceivers/
File: array.kicad_sch

Title: Transceivers: 64 channels (8 x 8 grid)

Size: A3	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 8/15

 This sheet is used 64 times; the page shows the first instance.

Transceiver channel (one of 64). TX: UCC27524 half -> 47 R -> BAV99 crossed diodes -> element (0–15 V square).
33k bleed resets the element after TX. RX: 4.7k + BAV99 clamp (+–0.7 V) + 100 pF (340 kHz RF filter).
OPA1678 stage 1: gain 6, AC-coupled at 800 Hz. Stage 2: MFB band-pass f0 39.8 kHz, Q 2, gain –4. Then 100 R to the ADC.
Element -> amplifier out +27 dB (x22), -> ADC pin +16.5 dB after the ADC cell's 2k/1k divider; pass band 31–51 kHz (sim/reports/rx_chain.md). Band-pass parts 1 % C0G.



Hybrid 8x8 ultrasonic AESA

Sheet: /Transceivers/CH01/
File: channel.kicad_sch

Title: Transceiver channel (x64): TX, T/R switch, RX amplifier

Size: A4	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 9/15

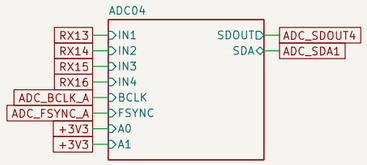
<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

ADCs. ADC k digitises RX 4(k−1)+1 … 4k and returns one TDM line (SDOUTk). Group g (ADC 4g−3 … 4g) shares SDAg and BCLK/FSYNC output g; A1/A0 give each ADC of a group its own I2C address.

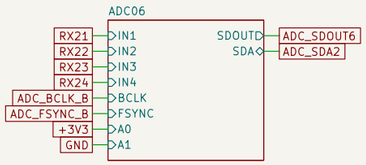


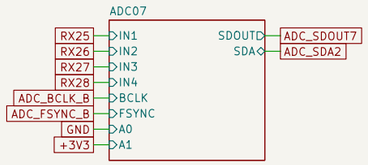


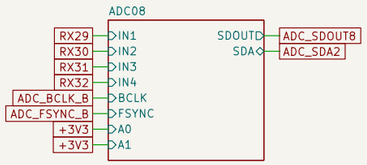














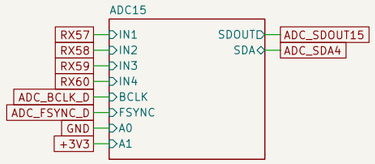


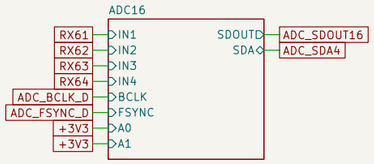












Hybrid 8x8 ultrasonic AESA

Sheet: /ADCs/
File: adcs.kicad_sch

Title: ADCs: 16 x TLV320ADC5140 (4 channels each)

Size: A3	Date: 2026−10−24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 10/15

ADC: TLV320ADC5140 (TI, 4 channels, up to 768 kSPS). Run at 192 kSPS: flat to about 86 kHz.

TDM slave: BCLK 18.432 MHz, FSYNC 192 kHz from the FPGA (shared by all 16 ADCs, so all 64 channels sample at the same instant). 4 slots x 24 bits on SDOUT.

I2C: SCL shared, SDA per group of 4 ADCs; A1/A0 set the address 0x4C–0x4F inside the group. Config writes can use I2C broadcast (P0_R2 I2C_BRDCAST_EN).

Per channel, set at boot: CHx_INSRC = single-ended (01), CHx_IMP = 10 k (01), AC-coupled, fixed 0 dB gain, DRE and AGC OFF (they change gain on the fly and break channel matching).

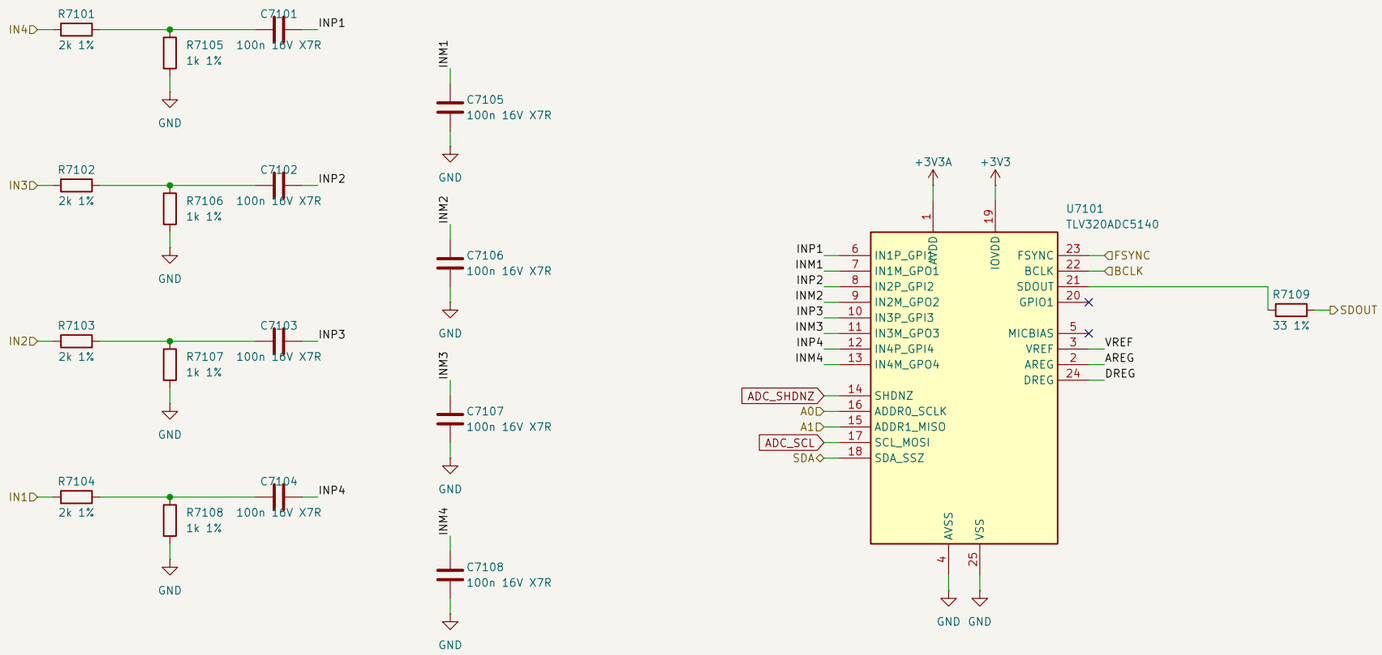
Also: ADC_FSCALE 2.75 V (1 Vrms single-ended full scale), AREG_SELECT = 1, TX_EDGE = 1 (the FPGA latches SDOUT one BCLK after launch).

Input: 2k/1k divider (x0.31 with the 10 k input) keeps the receiver's +-4.2 V worst case inside the 1.41 V peak full scale; 100 nF couples to the ADC's own bias (150 Hz high-pass).

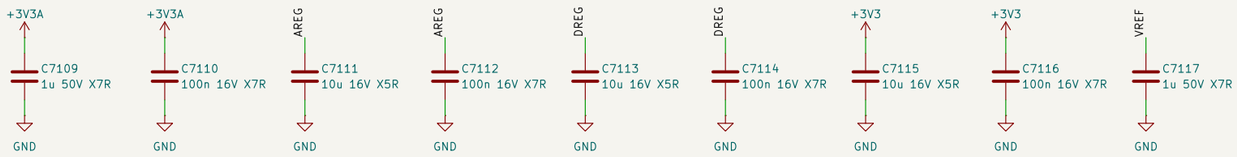
Power-up: SHDNZ (ADC_SHDNZ, pulled low on the header sheet) stays low until the rails settle, then wait 2 ms before I2C.

Channel inputs: 2k + 1k divider, 100 nF AC coupling (single-ended, INxM AC-grounded)

Ports IN1–IN4 are the block's four elements in order; IN1 feeds ADC channel 4 ... IN4 channel 1 (the order the receive lanes reach the inputs on the board). TDM: channel c carries element 4k-4 + (5 − c).



Decoupling at the pins (datasheet Figure 172): AVDD 1 uF + 100 nF, AREG and DREG 10 uF + 100 nF, IOVDD 10 uF + 100 nF, VREF 1 uF



Hybrid 8x8 ultrasonic AESA

Sheet: /ADCs/ADC01/

File: adc.kicad_sch

Title: ADC (x16): TLV320ADC5140, 4 channels, 192 kSPS TDM

Size: A3 | Date: 2026-10-24 | Rev: 1

KiCad E.D.A. 10.0.6 | Id: 11/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

MEMS mics. Mics 2k−1 and 2k share MIC_DATAk−1 (SEL high on the odd one). One clock per mic pair: MIC_CLK_A mics 1−2, B 3−4, C 5−6, D 7−8.

















Hybrid 8x8 ultrasonic AESA

Sheet: /MEMS mics/
File: mics.kicad_sch

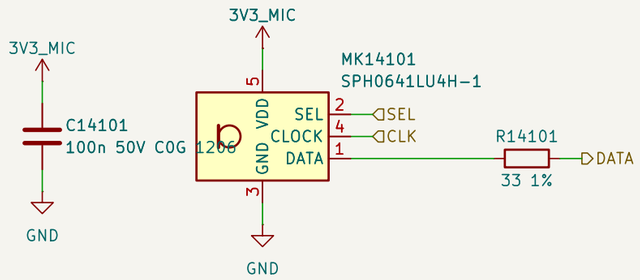
Title: MEMS microphones: 8 x SPH0641LU4H−1

Size: A3	Date: 2026−10−24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 12/15

<< up to: MEMS microphones: 8 x SPH0641LU4H-1 This sheet is used 8 times; the page shows the first instance.

MEMS mic SPH0641LU4H-1 (PDM). Two mics share DATA: SEL high on one, low on the other.

Mounted on the back, sound port through a board hole. 3V3_MIC is the ferrite-filtered rail (power sheet).

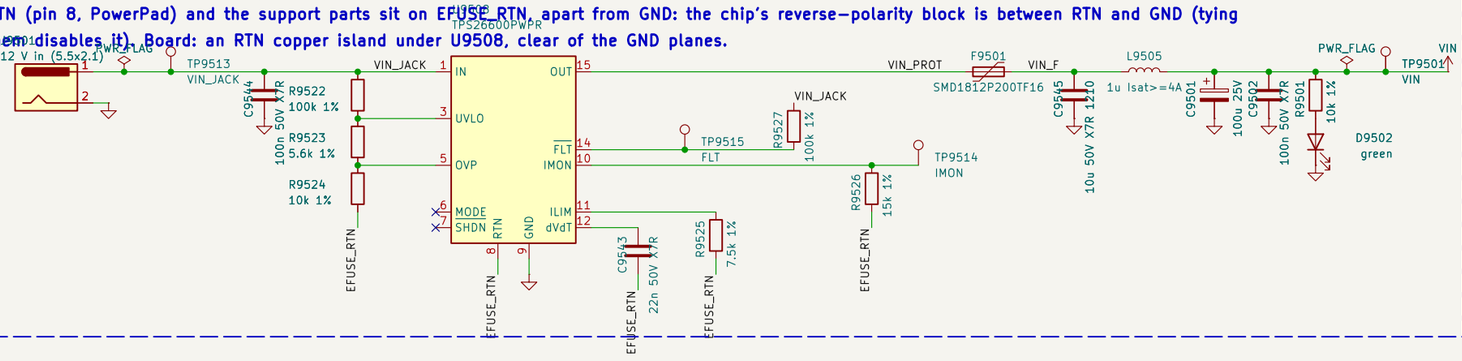


Hybrid 8x8 ultrasonic AESA		
Sheet: /MEMS mics/MIC1/		
File: mic.kicad_sch		
Title: MEMS mic (x8): SPH0641LU4H-1		
Size: A4	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 13/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

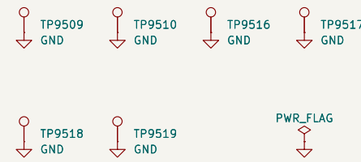
1. 12 V INPUT: jack J9501 -> TPS26600 eFuse U9508 -> polyfuse F9501 -> VIN

eFuse U9508 (resettable, auto-retry): rated +-60 V (reverse polarity), on above 8.8 V, off above 13.8 V (LM2731 input limit 14 V), current limit 1.6 A. Polyfuse F9501 (2 A) backs it up. C9545 - L9505 - C9501 keep the switchers' ripple off the cable.
UVLO/OVP divider R9522 / R9523 / R9524 (100k / 5.6k / 10k). ILIM R9525 7.5k = 1.6 A. dVdT C9543 22 nF: 5 V/ms soft start. IMON R9526 15k = 1.08 V/A (TP9514). FLT low = fault (TP9515). MODE open = auto-retry, SHDN open = on.
RTN (pin 8, PowerPad) and the support parts sit on EFUSE_RTN, apart from GND: the chip's reverse-polarity block is between RTN and GND (tying them disables it). Board: an RTN copper island under U9508, clear of the GND planes.



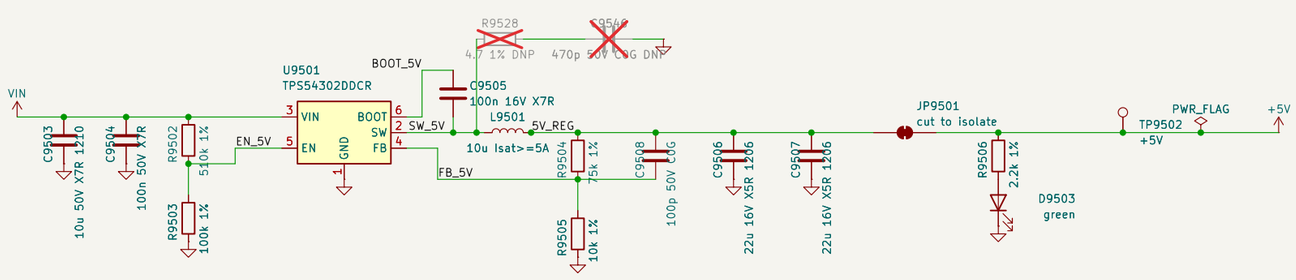
GROUND TEST POINTS

One GND net and plane (inner layers In1, In2). Scope-ground test points: TP9509 / TP9510 at the power supply, TP9516 - TP9519 spread over the board (ADC row, both array sides, TX row). Power tree and bring-up checklist: Power 2.



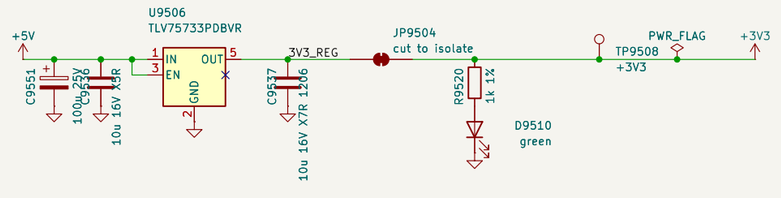
2. +5V: TPS54302 buck U9501 (TI reference design), 3 A

Vout = 0.596 V x (1 + 75k/10k) = 5.07 V (R9504 / R9505). EN UVLO about 7.3 V (R9502 / R9503). Load about 1.6 A. The switching loop C9503 - U9501 - L9501 is kept tight on the board. R9528 + C9546: switch-node snubber, DNP (fit only if the switch node rings).



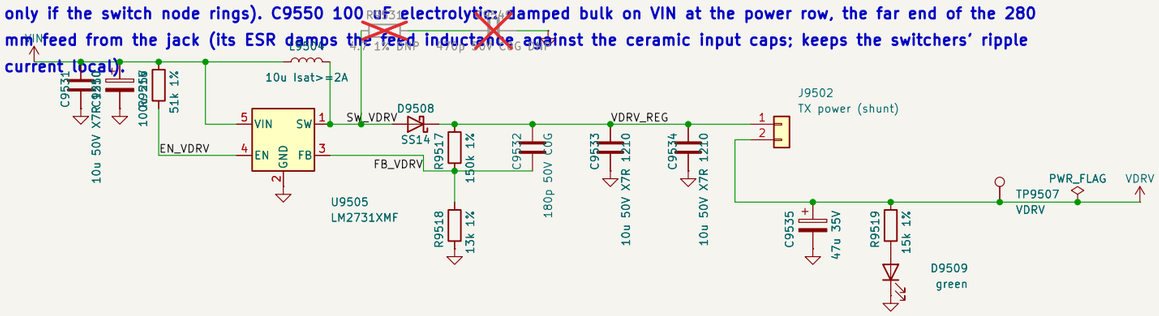
3. +3V3: TLV75733P LDO U9506 from +5V

1 A, stable with ceramic caps. Load < 0.15 A: ADC IOVDD, clock buffers, logic, mics.
C9551 100 uF electrolytic: damped bulk at the end of the 300 mm +5V run from the buck, for both top LDOs (U9506, U9507).



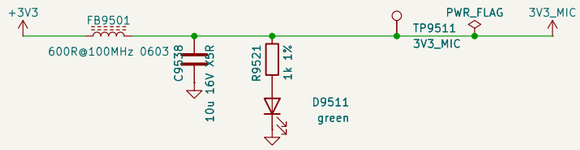
4. VDRV 15 V: LM2731X boost U9505 from VIN

1.23 V x (1 + 150k/13k) = 15.4 V (R9517 / R9518). Load, all 64 elements in CW at 40 kHz: about 0.24 A (sim/reports/tx_cw.md). Remove the J9502 shunt to unpower every TX driver. R9531 + C9549: switch-node snubber, DNP (fit only if the switch node rings). C9550 100 uF electrolytic: damped bulk on VIN at the power row, the far end of the 280 mm feed from the jack (its ESR damps the feed inductance against the ceramic input caps; keeps the switchers' ripple current local).



5. 3V3_MIC: mic supply

Ferrite-filtered +3V3 for the 8 MEMS mics. Remove FB9501 to isolate.



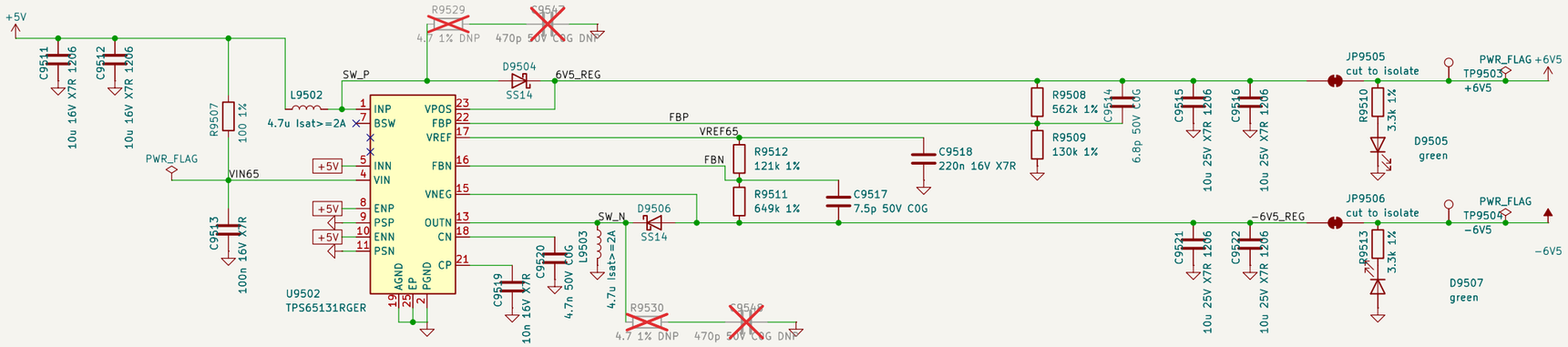
Hybrid 8x8 ultrasonic AESA
Sheet: /Power 1/
File: power.kicad_sch
Title: Power 1: 12 V input, +5V, VDRV (TX), +3V3, 3V3_MIC
Size: A3 | Date: 2026-10-24 | Rev: 1
KiCad E.D.A. 10.0.6 | Id: 14/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

6. +−6V5: TPS65131RGER U9502, boost + inverter from +5V (feeds the analog LDOs only)

Drawn with the pin-identical TPS65130 symbol: order TPS65131RGER (2 A switches). PSP/PSN low: fixed-frequency PWM. R9529 + C9547 and R9530 + C9548: switch-node snubbers, DNP (fit only if a switch node rings).

+6V5 = 1.213 V x (1 + 562k/130k) = 6.46 V (R9508 / R9509). −6V5 = −1.213 V x 649k/121k = −6.51 V (R9511 / R9512).



POWER TREE, BRING-UP

Power 1: sections 1–5 (input, +5V, +3V3, VDRV, 3V3_MIC) and the ground test points. Power 2: sections 6–8 (+−6V5, +−5VA, +3V3A).

1. 12 V jack J9501 -> eFuse U9508 -> F9501 -> VIN.
2. VIN -> buck U9501 -> +5V.
4. VIN -> boost U9505 -> VDRV.
3. +5V -> LDO U9506 -> +3V3 -> 5. FB9501 -> 3V3_MIC.
6. +5V -> U9502 -> +−6V5 -> 7. LDOs U9503 / U9504 -> +−5VA (receivers).
8. +5V -> LDO U9507 -> +3V3A (ADC AVDD).

First power-up: J9502 shunt off, bench supply 12 V limited to 0.3 A. Expect at the test points:

VIN 12 V, +5V 5.07 V, +3V3 3.30 V, 3V3_MIC 3.3 V, +6V5 6.46 V, −6V5 −6.51 V, +5VA 5.00 V, −5VA −5.00 V, +3V3A 3.30 V.

IMON (TP9514) 1.08 V per A in. FLT (TP9515) high (low = eFuse fault).

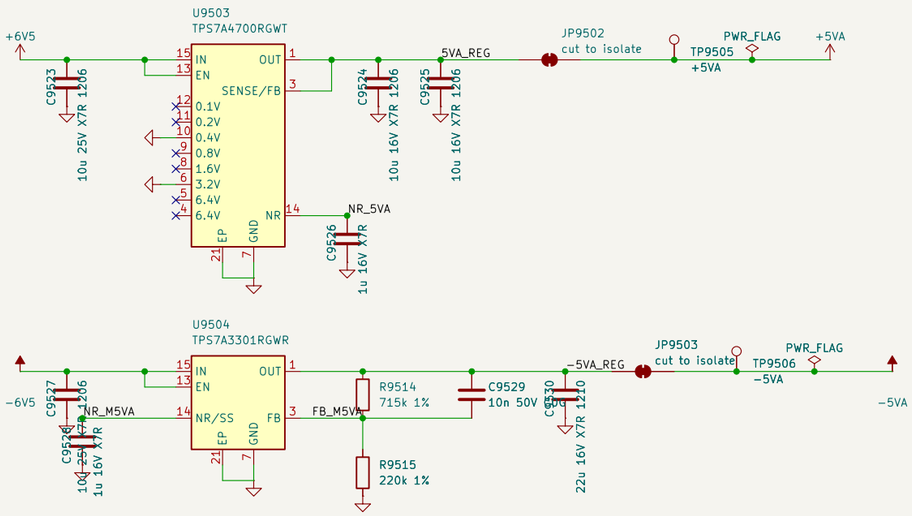
Then fit the J9502 shunt: VDRV 15.4 V.

A wrong rail: cut its jumper, measure the regulator side (*_REG) and the load side apart.

7. +−5VA: low-noise LDOs TPS7A4700 U9503 / TPS7A3301 U9504

+5VA: ANY-OUT 1.4 + 3.2 + 0.4 = 5.0 V. −5VA = −1.176 V x (1 + 715k/220k) = −5.00 V.

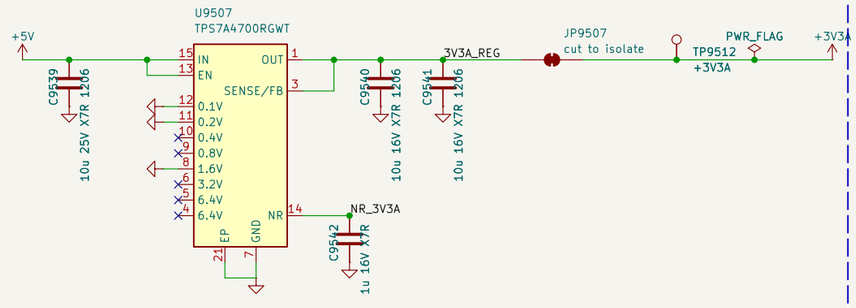
Load: 64 x OPA1678 (4 mA per channel), about 0.3 A on each rail. Analog rails: no LEDs.



8. +3V3A: TPS7A4700 low-noise LDO U9507 from +5V

AVDD of the 16 ADCs (analog rail: no LED). ANY-OUT 1.4 + 1.6 + 0.2 + 0.1 = 3.3 V.

Load about 0.45 A; 0.8 W in the LDO, thermal vias under the pad.



Hybrid 8x8 ultrasonic AESA

Sheet: /Power 2 (analog)/

File: power_analog.kicad_sch

Title: Power 2 (analog): +−6V5, +−5VA, +3V3A

Size: A3	Date: 2026-10-24	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 15/15